ACADEMIC INTELLIGENCE SYSTEM

Early Warning Agent

A multi-agent agentic system that continuously monitors student academic signals, synthesises evidence across seven specialist domains, and surfaces risk-stratified findings to human academic advisors before students reach the point of no return.

System Type	Agentic QA-RAG Pipeline
Institution	University / HEI
Privacy Framework	POPIA · FERPA-aligned
Decision Model	Human-in-the-Loop
Risk Levels	HIGH · MEDIUM · LOW · NONE

PROBLEM STATEMENT

Students at risk are invisible until it's too late

Most higher education institutions lack the infrastructure to detect academic risk proactively. The Academic REW Agent addresses this gap by synthesising signals from across a student's academic record into a single, evidence-backed risk profile — delivered to the right human, at the right time.

THE GAP

In most higher education institutions, academic risk is only detected reactively — after a student fails a module, misses an examination, or has already disengaged entirely. By the time a lecturer or advisor notices, meaningful intervention is no longer possible within the same semester.

THE COMPLEXITY

Student risk is not a single number. It is the intersection of academic performance trends, attendance patterns, module-level difficulty, psychosocial stressors, financial pressures, and historical trajectories — data spread across multiple disconnected institutional systems.

THE SCALE

A single lecturer may be responsible for hundreds of students across several modules. A tutor carries a caseload that makes individual monitoring impossible without intelligent assistance. Human attention is finite; risk does not distribute itself conveniently.

THE GOAL

The Academic REW Agent provides a continuously updated, evidence-backed, risk-stratified view of every student — synthesised automatically from all available signals — so academic staff can focus their limited intervention capacity on the students who need it most, at the right moment.

SYSTEM ARCHITECTURE

Brain, Perception, and the Flow of Evidence

The diagram below maps the complete architecture of the Academic REW Agent. Data enters from the Perception layer (left), passes through the Agentic Brain — comprising the Academic Knowledge Base, seven specialist review agents, and two quality-control agents — before reaching the human decision gate. The Learning Element (right) captures intervention outcomes and retrains the system continuously.

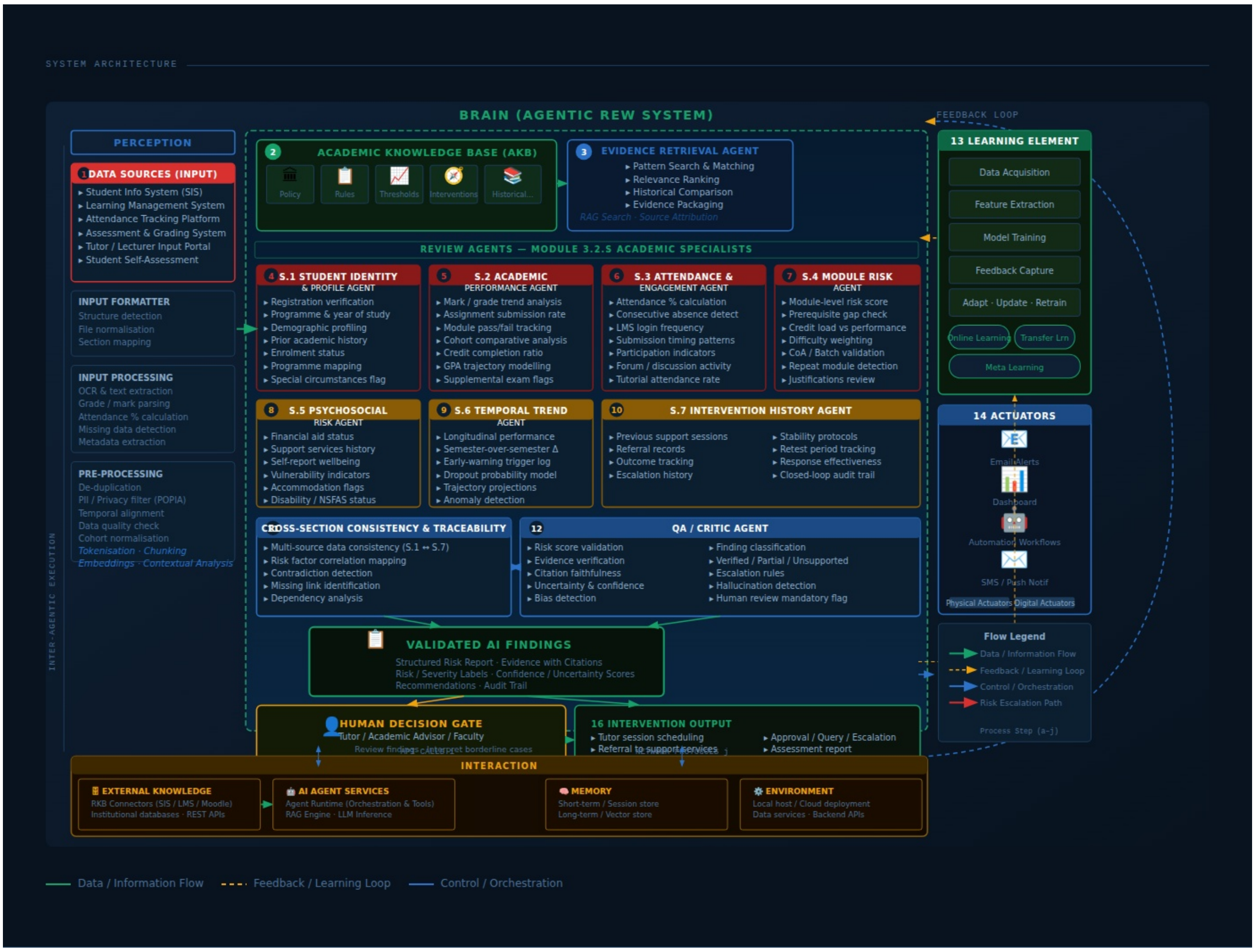

Figure 1 — Academic REW Agent: Full system architecture showing Perception, Brain (Agentic QA-RAG System), Learning Element, Actuators, and Interaction layers.

HOW THE AGENT WORKS

A seven-layer intelligence pipeline, closed by human judgement

The Academic REW Agent operates as a continuous, multi-stage agentic loop. Data enters through the Perception layer, is enriched and embedded, processed by seven specialist agents in parallel, reconciled by consistency and QA agents, and delivered as structured findings to the academic professional who holds the decision authority. The system never acts autonomously — every intervention is approved by a human.

PHASE 01

Data Ingestion & Normalisation

Student records are pulled from the SIS, LMS, attendance and grading platforms via REST APIs and scheduled data feeds. The Input Formatter detects document structure, normalises file formats, and maps fields to a canonical academic schema. PII filtering is applied under POPIA compliance before any embedding or analysis. Missing data is flagged explicitly rather than silently interpolated.

PHASE 02

Knowledge Base Query

The Academic Knowledge Base (AKB) holds institutional policy, programme rules, risk threshold configurations, and historical cohort patterns. The Evidence Retrieval Agent runs RAG search across the AKB to retrieve relevant precedents — students with similar profiles, historical intervention outcomes, and module-specific risk baselines.

PHASE 03

Parallel Specialist Analysis

Seven specialist agents run concurrently across their assigned domains: identity & profile, academic performance, attendance & engagement, module risk, psychosocial risk, temporal trend modelling, and intervention history. Each agent produces a structured partial finding with a confidence score and severity label. No single agent can produce a final risk score — synthesis is always required across domains.

PHASE 04

Cross-Section Reconciliation

The Consistency & Traceability Agent maps all seven specialist outputs, checking that risk signals are internally coherent and that no critical data dependency is broken. Contradictions are surfaced explicitly rather than averaged away. A full traceability graph links every risk claim to its source data and the agent that generated it.

PHASE 05

QA & Critic Validation

The QA / Critic Agent independently verifies each finding: is the evidence sufficient? Is the claim proportionate? Is there a hallucination or unsupported inference? Findings are classified as Verified, Partially Supported, Unsupported, or Human Review Mandatory. The final risk score is computed only after QA passes, incorporating confidence weights from each specialist domain.

PHASE 06

Human Decision Gate

Validated findings are presented to the tutor, lecturer, or academic advisor — never acted on autonomously. The professional reviews the evidence package, interprets borderline and complex cases the agent has flagged, and makes the final intervention decision. All decisions are documented, creating a closed-loop audit trail the institution can rely on for compliance.

PHASE 07

Intervention & Feedback

Approved interventions are dispatched via the Actuator layer: tutor session bookings, email alerts, referrals to student support services, or formal academic plan adjustments. Intervention outcomes are fed back into the Learning Element, which retrains and recalibrates risk models based on real-world results. The system improves with each cohort.

SPECIALIST AGENT REGISTER

Seven agents, one integrated risk picture

Each specialist agent is scoped to a single domain of student risk. They run in parallel, cite sources independently, and are never allowed to override one another — synthesis always happens at the reconciliation layer above. The QA / Critic Agent provides an independent validation pass before any finding reaches a human decision-maker.

AGENT	NAME	RESPONSIBILITIES
S.1	**Student Identity & Profile**	Registration & enrolment verification; Programme mapping & year of study; Prior academic history & RPL; Special circumstances flags; Demographic profiling (POPIA)
S.2	**Academic Performance**	Mark / grade trend analysis; Assignment submission rate; Module pass/fail & supplemental history; Cohort comparative benchmark; GPA trajectory modelling
S.3	**Attendance & Engagement**	Attendance % per module; Consecutive absence detection; LMS login frequency; Submission timing pattern analysis; Forum participation indicators
S.4	**Module Risk**	Module-level risk scoring; Prerequisite gap identification; Credit load vs. performance ratio; Module difficulty weighting; Repeat module detection
S.5	**Psychosocial Risk**	Financial aid & NSFAS status; Student support services history; Self-reported wellbeing; Disability / accommodation status; Vulnerability flag integration
S.6	**Temporal Trend**	Longitudinal performance curves; Semester-over-semester delta; Early-warning trigger timeline; Dropout probability modelling; Anomaly detection across time series
S.7	**Intervention History**	Previous support sessions & outcomes; Referral records; Escalation history; Response effectiveness scoring; Closed-loop audit trail
QA	**QA / Critic Agent**	Risk score validation; Evidence sufficiency verification; Hallucination & bias detection; Finding classification (Verified / Partial / Unsupported); Human Review Mandatory flagging

SECURITY & COMPLIANCE

Built for institutional trust

POPIA Compliance

All student data is filtered for PII before embedding or analysis. No personally identifiable information is stored in the vector knowledge base.

Row-Level Security

Backend access is enforced through Supabase RLS. Service-role keys are never exposed to the frontend. Each user role (student, tutor, lecturer, admin) sees only the data they are authorised to access.

Human-in-the-Loop

The system never acts on a student without explicit human approval. Every AI finding is labelled with a confidence score and classified as Verified, Partially Supported, Unsupported, or Human Review Mandatory.

Audit Trail

Every decision — including 'no action' — is documented with timestamp, actor, rationale, and the AI findings that informed it. The trail is immutable and available for institutional review and regulatory inspection.

Production Safeguards

Developer testing tools are disabled in production environments. No test credentials or bypass mechanisms persist to the production deployment.